To be FAIR,
theory needs an update!

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Context

- Open science practices have improved hypothesis testing,
  but not **theory development.**
- Psychological theories are often **ambiguous**, **hard to test**, **rarely updated**, and **not integrated into scientific workflows**
- **FAIR theory** addresses these shortcomings

What is FAIR Theory?

- **Findable** in indexed FAIR-compliant repositories with DOI and metadata.
- **Accessible:** Plain-text, readable by humans and machines.
- **Interoperable** for specific purposes like hypothesis derivation, covariate selection, data simulation, and iterative improvement.
  - Can be interoperable in analysis software (R-package theorytools)
- **Reusable:** Iteratively improvable by the broader community through clear licensing and version control.

How?

- **Specify theory** clearly as prose, formula, diagram, or other (formal?) model.
- **Archive theory object** in open repositories (e.g., Zenodo) with appropriate metadata.
- **Version-control** theory development using Git and GitHub.
- Use **semantic versioning** (MAJOR.MINOR.PATCH) to communicate changes to users.

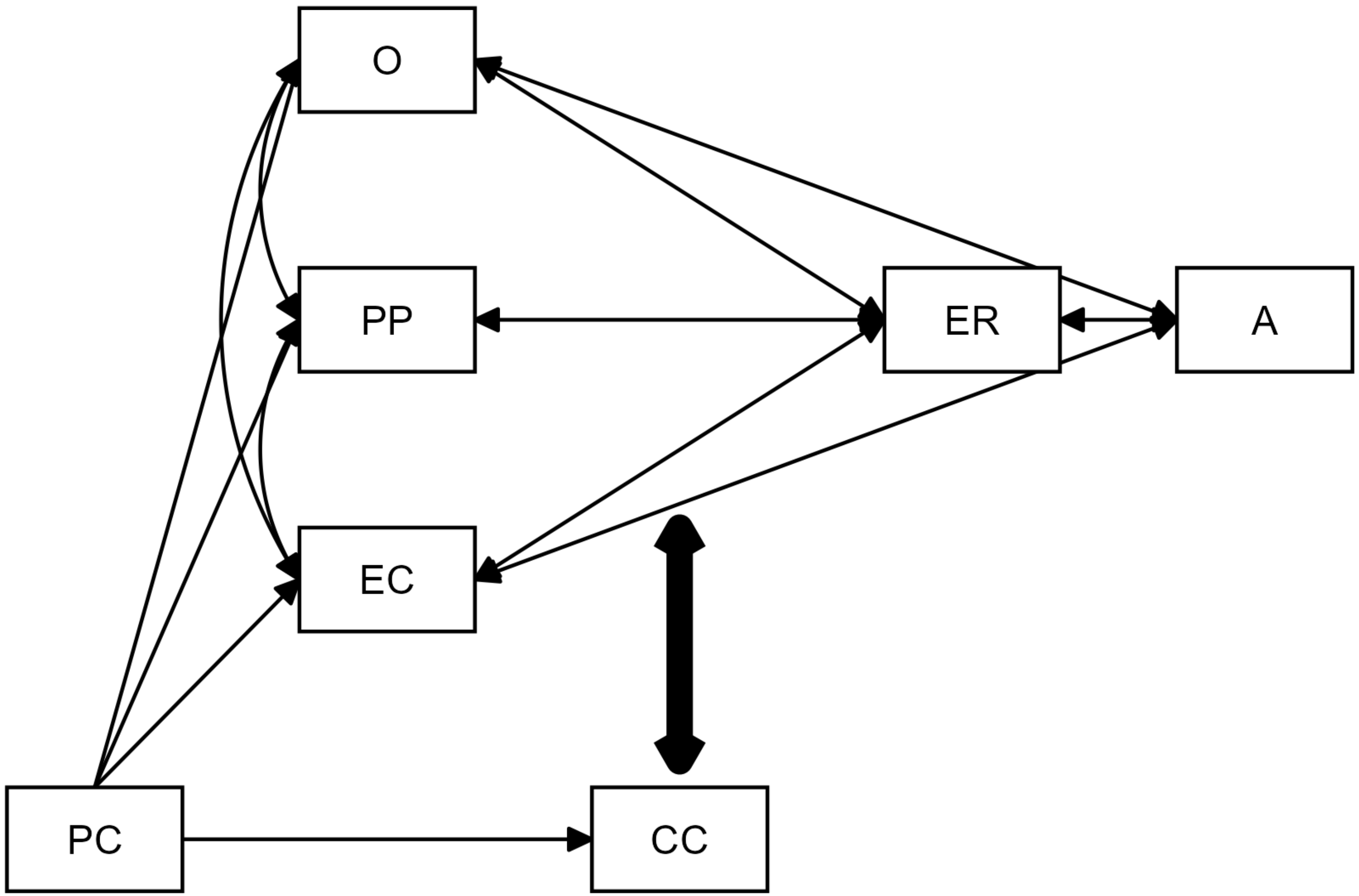
Why Make Theory FAIR?

- **Reduces research waste**; theories are shared, reused, and improved.
- **Enables meta-research** on structure and evolution of theories.
- Enables **analogical modeling**: applying structure of a theory from one field to problem from another.
- Integrates theory into **reproducible research workflows** (`worcs`, `theorytools`).
- **Accelerates cumulative knowledge acquisition** in psychology and beyond.

Impact

- Supports **modular publishing**: standalone, citable theory artifact.
- Aligns with **open science principles** and new **research evaluation standards** (e.g., DORA, CoARA).
- Facilitates **team science**, interdisciplinary work, and evidence-based decision-making.
- Promotes **theoretically sound** research.
- **Aids science communication: theories**

Example: Morris' Tripartite Model

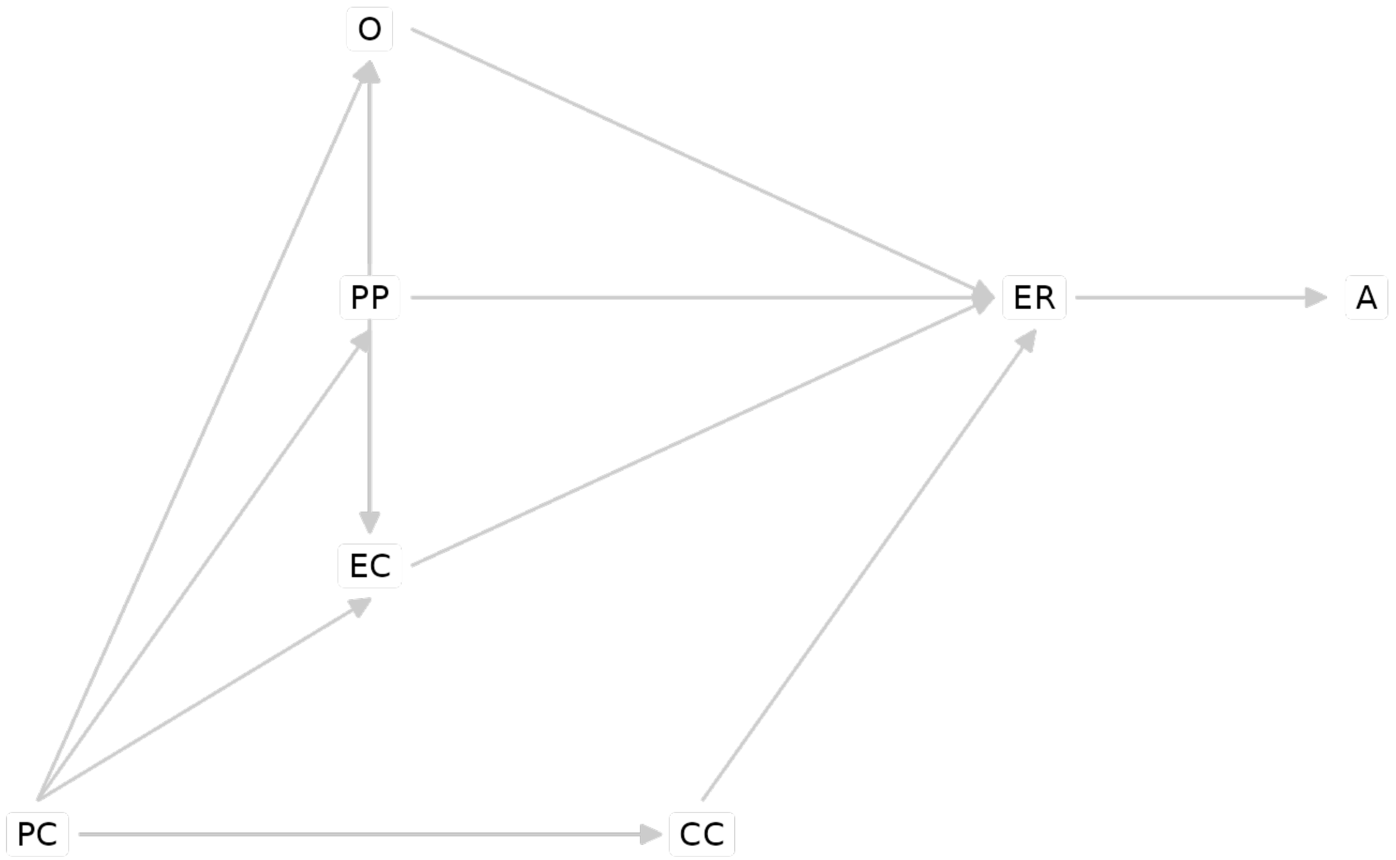


Tripartite Model as FAIR Theory

```
tripartite <- dagitty('dag
{

PC -> CC

PC -> EC

PC -> PP

PC -> O

O -> ER

PP -> ER

EC -> ER

ER -> A

CC -> ER
[form="CC:O+CC:PP+CC:EC"];

PP -> O

EC -> O

PP -> EC

}')
```

Interoperability: Downloading and Plotting

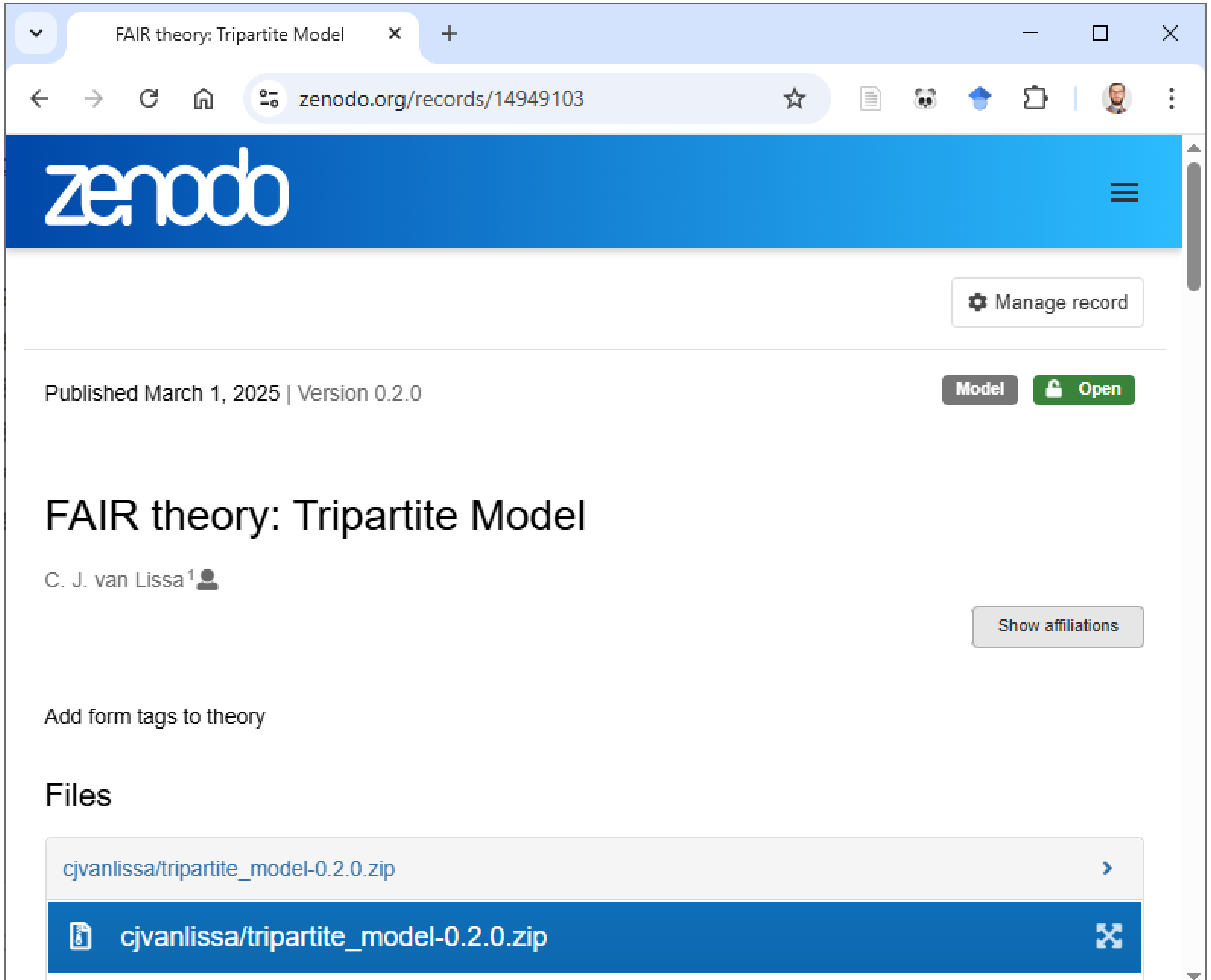
```
download_theory(https://doi.org/10.5281/zenodo.14921521)
graph_sem(tripartite)
```



Interoperability: In Analysis Workflow

```
# Simulate data
df_sim <- simulate_data(tripartite, n = 497)

# Power analysis with hypothesized parameter values
tripartite_coef <- dagitty('dag {
PC -> CC [beta=.4]
PC -> EC [beta=.2]
PC -> O [beta=0]}')

# Select covariates for causal inference
adjustmentSets(tripartite, exposure="O", outcome="ER")
#> { CC, EC, PP }
#> { EC, PC, PP }
```

Archived: On Zenodo



Discussion

- Discuss with your neighbor: What is a “theory” in your field?

Discussion

- Discuss with your neighbor: How FAIR are theories in your field? Give good/bad examples

Discussion

- What are/would be the most important benefits of FAIR theory for you / your field / society?

Discuss

- How are new theories made?
- Popper: “there is no such thing as a logical method of having new ideas, or a logical reconstruction of this process.”
- What would your method be?

Exercise

- Take one specific theory in your field (or one that most of us would know)

- What would need to happen to make it FAIR?

- Is this theory regularly updated and improved? Could / should / how would it be updated?